Current FD route: nine-category source coverage review

Task T-1564. 3 November 2026.

Historical joins	10 exact files; 7 reviewed deltas
FD source	Degree2: -5p M; 2-2p adds; 5-5p visits
Nine categories	Source coverage accepted in finite model
Next binding	Root integration; executable / protocol / image

Ordinary software evidence only. Source review; no compiler or runtime. T-1564.

The selected diagnostic finite-difference route retains coverage of all nine categories in the declared finite algorithm model at the reviewed source. Fresh comparison verifies ten of seventeen historical files byte-for-byte; seven changed files were reviewed by delta. A direct diff of the historical solver against the current progress module isolates the FD replacement and progress-observation hooks. This is a source acceptance, with final executable, protocol and image bindings pending.

Reviewer witness_independent_review_v2 did not author the historical pricing policy, mapper, FD producer or progress implementation. Prior context came from writing the exporter and build-binding plans. This is independent reviewer scrutiny with nonblind integration context, not a newly independent algorithm. PLAN.md was frozen before detailed inspection; published revision 5ea0e1e75a93ef025fb6f25645a7936ad06ccb9d includes the applied route revision 2d17cb240d40de09f58cb92f8f8a37d04337b575. No compiler, scientific payload, cloud action or trial was run.

For degree2, the old evaluation read five coefficients, performed five multiplications and five additions at each of p points. The new recurrence reads five coefficients once, initializes two additions, and performs three additions at every point, including the last iteration and before both continue branches. Hence exact candidate-minus-baseline changes are -5p multiplications, 2-2p additions and 5-5p collection visits. At p=19 these are -95, -36 and -90. This is a hand derivation of source operations, not a new measured trial. For general supported degree d the old coefficient count is 2d+1; the selected production caller only invokes the sieve for degree2.

The recurrence uses u(x+1)=u(x)+delta(x), delta(x+1)=delta(x)+2u2 and v(x+1)=v(x)+v1. All updates invoke the existing charged add helper. Canonical coefficients and 0<p<=2^63 make a+b <=2p-2 fit u64. The source documents this bound but does not add a public bound guard; acceptance is restricted to the intended finite prime cohort, not all possible u64 input. Inactive coefficients are ignored. Exceptional roots, empty rational-y branches and batch inversions retain their historical operations and counters.

Field/scalar arithmetic, primality/constructor work, setup/transfer, graph/collection/elimination, hash/RNG, source verification, metadata clones and calibration are mapped in categories.csv and review.json. The actual mapper arithmetic is unchanged. It removes the inversion loop expansion exactly once before applying an atomic inverse charge; the loop includes its last square. Setup field/native arithmetic is already in candidate totals; setup symbolic counts enter separately. Geometry totals are partition checks, not extra operations to add over their segments. Calibration preparation plus addition equals total, and method-only and combined prices remain separate.

The adapter chooses Some(sink) progress_native explicitly and retains graph owners through verification and reporting. Target-independent applicability now occurs before the timer and geometry snapshot. Its moved parameter/family predicates invoke no selected arithmetic or geometry observers; this preserves selected event coverage while changing the wall boundary. Actual record import still requires a non-null session despite nullable planned identities. Reporting-only library exports and test-only rho hash changes do not alter the selected production event map.

Progress snapshots use plain serialization and the concrete AtomicSink, without invoking algorithm counters. Their CPU, bytes, allocations and filesystem work remain distinct resource costs. The sink's completed_write_ns does not cover every snapshot construction operation. Partial snapshots remain diagnostic-only and cannot establish an accepted result. The canonical optional Candidate relation trace is absent from the progress module, so this route does not supply a full retained relation stream.

No additional missing modeled source event was found in the reviewed deltas. Existing exact historical sources are reused only at matching hashes; changed-file reasoning is separately retained. This report is not an executable coverage admission record and deliberately leaves binary/protocol/image identities null. Full root adapter integration, final worker/mapper builds and source closures, matched diagnostic controls, and retained run custody remain concrete downstream requirements. The historical qualification and price records are unchanged.